Ayesha Andleeb

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Honors & Achievements

- Selected as Section Leader(Trainer) at **Stanford University: Code In Place**. **[link]**
- **Merit Scholarship Recipient** in Senior Year for a perfect 4.00/4.00 semester GPA.
- 7X International Hackathon **Mentor & Judge**
- 1st place Winner of **Harvard** CS50x Puzzle Day by solving 9/9 Algorithmatic puzzles**.** **[link]**
- Invited as a **guest speaker** for a webinar on the "Generative AI Essentials" to 300 + students. **[link]**
- Ranked in the top 12% among 820+ international teams at the **UC Berkeley** Coding Competition. **[link]**
- Qualified for **Meta Hacker Cup** Round 1, ranked 81st globally & **5th in Pakistan.** **[link]**
- Secured **3rd position** out of 120+ international teams at Codestral Hackathon. **[link]**
- Completed 50/50 challenges in the 25-day **Advent of Code 2024** series, secured **46th national rank**. **[link]**

EDUCATION

University of Sahiwal — Sahiwal, Pakistan

Bachelor of Science in Information Technology, **CGPA: 3.58/4.00** — Dec 2021 - Jun 2025

Final Year Project: Deep Learning-Based Brain Tumor Classification System | [link]

Tools: Python, TensorFlow, Keras, NumPy, Pandas, Matplotlib, Kaggle Dataset

- Developed a four-class brain tumor MRI classifier using EfficientNetB0 transfer learning, achieving ~94% accuracy.
- Built a classification pipeline for glioma, meningioma, pituitary, and healthy tissue detection.
- Applied Grad-CAM Explainable AI to generate saliency heatmaps over MRI scans, verifying that model attention aligns with clinically relevant tumor regions to meet Explainable AI standards.
- Evaluated model performance using F1-score, ROC-AUC, and confusion matrices.

VOLUNTEER TEACHING EXPERIENCE

Stanford University: Code In Place (Remote) — CA, USA

Role: Section Leader (trainer) — Apr-Jun

- Selected as a Trainer for **3 consecutive years** (2024, 2025, 2026) from a global pool of **2,000+ applicants** each year.
- Led 6 weekly live Python sessions for 12+ international students, using Karel-based problem-solving.

Pak Angels (Remote) — CA, USA

Role: Co-Lead Instructor, Mentor & Hackathon Judge — Jun -Jul 2026

- Delivered a 6-week GenAI curriculum covering LLM APIs, RAG pipelines, and prompt engineering to 1,000+ students
- Evaluated **100+ final AI projects** as a hackathon **mentor & judge**.

Iqra Fund (Remote) — NV, USA

Role: Python Instructor — Dec 2024 - Jan 2025

- Taught **Python fundamentals**, including variables, loops, functions, and debugging.
- Led interactive coding sessions using basic Microsoft tools for over **70+ underprivileged girls.**

iCodeGuru (Remote) — CA, USA

Role: Course Trainer and Moderator — Jun 2023- Jul 2025

- Taught Python, Data Structures & Algorithms, and Generative AI to **500+ students**
- Through **100+ hours** of live, hands-on sessions.
- Delivered a 5-week IELTS/Duolingo prep course.
- Helped students build academic English skills for international university admissions.

WEBINARS AND WORKSHOPS

- Delivered a **2-hour** online workshop on **"How to Build Your First AI App from Idea to Deployment."** [link]
- Mentored a 5-day **"UC Berkeley Calico Contest"** prep workshops for **100+ students.** [link]
- Delivered an online workshop on **"Object Oriented Programming"** for **80+** underprivileged learners. [link]
- Invited as a guest speaker to share my teaching experience from Stanford University: Code In Place program. [link]
- Led a **Meta Hacker Cup** prep webinar, guiding participants on problem-solving and contest strategy. [link]

ACADEMIC PROJECTS

MediScan: Multi-Agent Retrieval-Augmented System for Clinical Report Analysis **[link]**

Tools: Python, LangChain, LLaMA-3.3 (Groq API), FAISS, Flask

- Built a multi-agent **RAG** system for medical report analysis across cardiology, pulmonology, and psychology domains.
- Developed a FAISS-based retrieval pipeline using clinical reference notes for diagnostic information retrieval.
- Evaluated routing consistency across medical domains and analyzed limitations in handling cross-domain conditions.

NLP-Based Phishing Website Detection System **[link]**

Tools: Python, Scikit-learn, NLTK, CountVectorizer, Logistic Regression, Streamlit, Pandas, NumPy

- Built an NLP-based phishing URL detection system using **549K+ labeled URLs** to classify websites as **phishing or malicious** or legitimate through lexical preprocessing and feature extraction.
- Trained and compared Logistic Regression and Multinomial Naive Bayes models, achieving 96.5% accuracy with strong phishing detection performance and low false-positive rates.
- Developed a Streamlit-based real-time phishing URL detection system by integrating preprocessing and machine learning classification into a single pipeline.

Computer Vision-powered Traffic Sign Recognition System **[link]**

Tools: Python, Keras, CNN, Flask, NumPy, Matplotlib

- Built a CNN-based system to classify 43 traffic sign categories under varying lighting and visibility conditions, achieving **~94% test accuracy** on unseen data.
- Applied class-weighted learning to improve recognition of underrepresented traffic sign classes.
- Integrated the model with Flask for real-time traffic sign detection with low-latency inference suitable.

INTERNATIONAL HACKATHON PROJECTS

Co-Creating with GPT-5 Hackathon (view details) Aug 2025

Project: SmartLearn (AI-Powered English Learning Assistant)

Tools: Python, Streamlit, FAISS(database), GPT-5 model, LangChain

- Created an AI tool that gives real-time **grammar help, vocabulary suggestions, and speaking practice feedback.**
- Added **personalized learning paths** so users get tasks based on their skill level.
- Used **semantic search** and LLM logic to give clear and natural explanations during practice.

Trae AI IDE: Zero Limits Hackathon (view details) Jun 2025

Project: GeneGazer (AI-Driven Single-Cell RNA-seq Analysis Platform)

Tools: Streamlit, Scanpy, Python, Google Gemini AI

- Built a platform to automatically process **single-cell RNA-seq data** and show interactive visual results.
- Applied steps like normalization, **PCA, UMAP,** and clustering for exploring cell-level data.
- Made the analysis faster and easier for users by adding guided workflows.

SKILLS

Programming Languages:	Python, JavaScript, SQL, Bash
Data Science Libraries:	TensorFlow, Keras, scikit-learn, NLTK, OpenCV, LangChain, FAISS, RAG pipelines
Tools & Deployment:	Git, Jupyter, Google Colab, Linux, Flask, Streamlit, REST APIs, Firebase